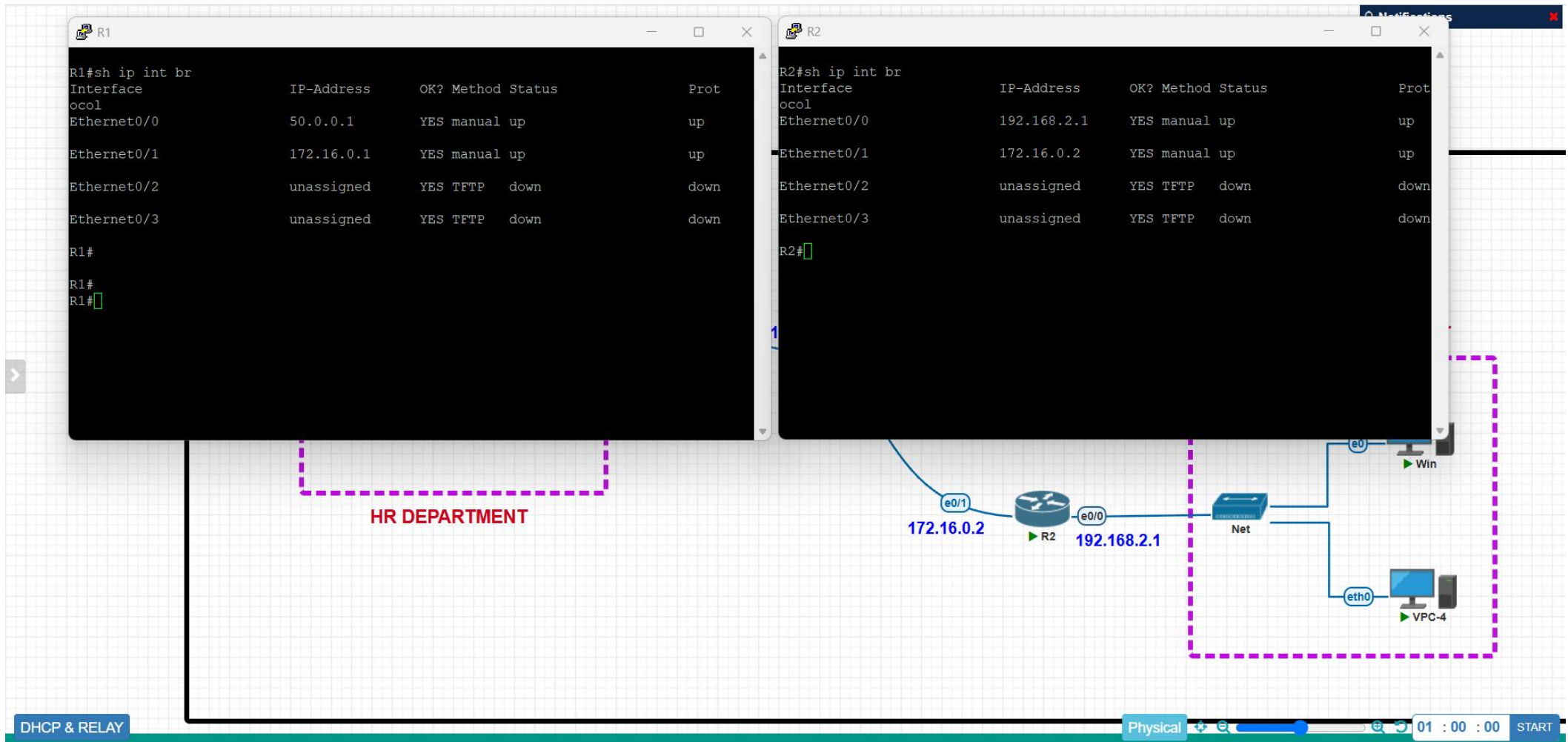


DHCP & RELAY LAB

ASSIGN THE IP ADDRESSES FOR ALL ROUTER 1 & ROUTER 2:-



ADD A STATIC ROUTE FOR MISSING ROUTES AT R1&R2:-

R1

```
R1#
R1# sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

  50.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       50.0.0.0/8 is directly connected, Ethernet0/0
L       50.0.0.1/32 is directly connected, Ethernet0/0
L       172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, Ethernet0/1
L       172.16.0.1/32 is directly connected, Ethernet0/1
S       192.168.2.0/24 [1/0] via 172.16.0.2
R1#
R1#
```

R2

```
Ethernet0/3          unassigned    YES TFTP    down          down

R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S       50.0.0.0/8 [1/0] via 172.16.0.1
L       172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, Ethernet0/1
L       172.16.0.2/32 is directly connected, Ethernet0/1
L       192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, Ethernet0/0
L       192.168.2.1/32 is directly connected, Ethernet0/0
R2#
R2#
```

HR DEPARTMENT

172.16.0.2

R2

192.168.2.1

Net

VPC-4

DHCP & RELAY

Physical

01 : 00 : 00

START

Air: Satisfactory
Now

Search

16:27
07-09-2026

ENABLE THE DHCP SERVER FOR HR AND SALES DEPT AT R1:-

labs.surenetwork.com/legacy/topology

Paused

```
R1#sh run
R1#sh running-config | section dhcp
ip dhcp excluded-address 50.0.0.1 50.0.0.50
ip dhcp excluded-address 192.168.2.1 192.168.2.50
ip dhcp pool HR
network 50.0.0.0 255.0.0.0
default-router 50.0.0.1
dns-server 8.8.8.8
lease 1 2 3
ip dhcp pool SALES
network 192.168.2.0 255.255.255.0
default-router 192.168.2.1
dns-server 8.8.8.8
lease 1 2 3
R1#
R1#
```

HR DEPARTMENT

SALES DEPARTMENT

172.16.0.2

192.168.2.1

Physical

01 : 00 : 00

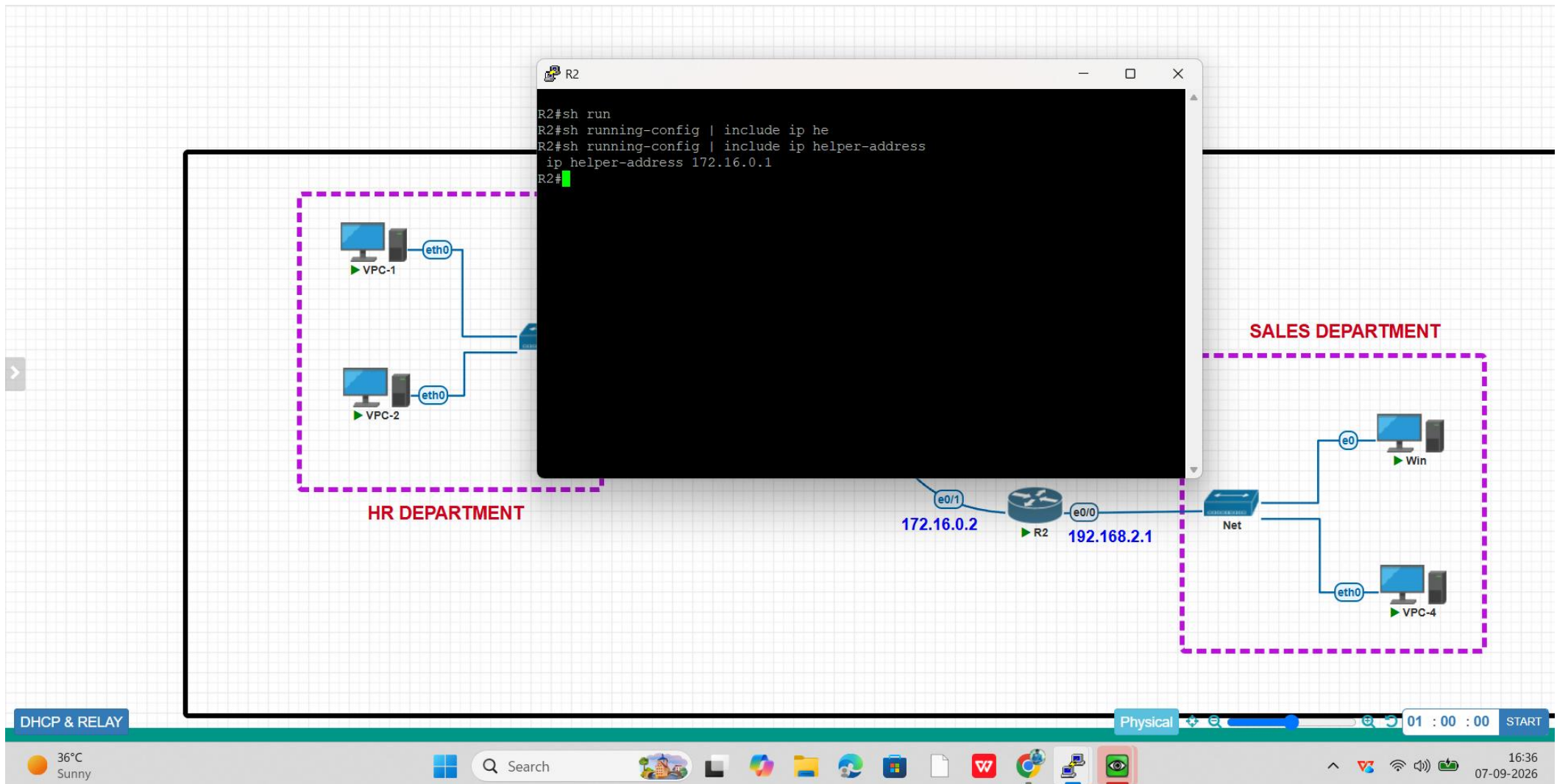
START

36°C Sunny

Search

16:31 07-09-2026

ENABLE DHCP RELAY AGENT AT R2:-



SEND A REQUEST FOR DHCP AND GET A IP FOR ALL THE PC'S:-

The screenshot displays a network simulation environment with three Virtual PC (VPC) windows and a network diagram.

VPC-1 (Top Left): Shows the Virtual PC Simulator interface. The command prompt displays the following text:

```
Welcome to Virtual PC Simulator, version 1.3 (0.8.1)
Dedicated to Daling.
Build time: May 7 2022 15:27:29
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
Copyright (c) 2021, Alain Degreffe (alain.degreffe@eve-ng.net)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version for EVE-NG.

Press '?' to get help.

Executing the startup file

VPC-1> ip dhcp
DDORA IP 50.0.0.51/8 GW 50.0.0.1

VPC-1> 
```

VPC-2 (Top Right): Shows the Virtual PC Simulator interface. The command prompt displays the following text:

```
Welcome to Virtual PC Simulator, version 1.3 (0.8.1)
Dedicated to Daling.
Build time: May 7 2022 15:27:29
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
Copyright (c) 2021, Alain Degreffe (alain.degreffe@eve-ng.net)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version for EVE-NG.

Press '?' to get help.

Executing the startup file

VPC-2> ip dhcp
DDORA IP 50.0.0.52/8 GW 50.0.0.1

VPC-2> 
```

VPC-4 (Bottom Center): Shows the Virtual PC Simulator interface. The command prompt displays the following text:

```
Welcome to Virtual PC Simulator, version 1.3 (0.8.1)
Dedicated to Daling.
Build time: May 7 2022 15:27:29
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
Copyright (c) 2021, Alain Degreffe (alain.degreffe@eve-ng.net)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version for EVE-NG.

Press '?' to get help.

Executing the startup file

VPC-4> ip dhcp
DDORA IP 192.168.2.52/24 GW 192.168.2.1

VPC-4> 
```

Network Diagram (Bottom Right): A network diagram showing a central router labeled "Net" connected to three devices: "VPC-2", "Win", and "VPC-4". The router's interface is labeled "e0". The "Win" device is connected to the router via "e0". The "VPC-4" device is connected to the router via "eth0". The diagram is enclosed in a dashed purple box.

Bottom Bar: A status bar at the bottom of the screen. On the left, it shows "DHCP & RELAY". In the center, it displays "Physical" and a progress bar. On the right, it shows a timer "01 : 00 : 00" and a "START" button. The system tray at the bottom indicates "36°C Sunny" and the date "07-09-2026".

WINDOWS PC DHCP IP OUTPUT:-

labs.surennetwork.com/legacy/topology

Paused

QEMU (Win)

```
Microsoft Windows XP [Version 5.1.2600.1]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 192.168.2.51
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.2.1

C:\Documents and Settings\user>
```

Local Area Connection Status

Network Connection Details

Property	Value
Physical Address	50-AE-34-09-11-00
IP Address	192.168.2.51
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1
DHCP Server	172.16.0.1
Lease Obtained	9/7/2026 11:26:20 AM
Lease Expires	9/8/2026 1:29:20 PM
DNS Server	8.8.8.8
WINS Server	

SALES DEPARTMENT

Net

Win

VPC-4

OPPO K13 5G 79E6
Internet access

00 START

36°C Sunny

Search

16:43 07-09-2026

DHCP DATA BASE AT R1:-

labs.surennetwork.com/legacy/topology

Paused

HR DEPARTMENT

VPC-1

VPC-2

SALES DEPARTMENT

Win

VPC-4

Net

R1

```
R1#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
Hardware address/
User name
192.168.2.51    0100.5079.6668.83   Sep 08 2026 05:51 PM    Automatic
192.168.2.52    0100.5079.6668.84   Sep 08 2026 05:53 PM    Automatic
192.168.2.51    0150.ae34.0911.00   Sep 08 2026 05:59 PM    Automatic
192.168.2.52    0100.5079.6668.85   Sep 08 2026 06:00 PM    Automatic
R1#
```

DHCP & RELAY

Physical

01 : 00 : 00

START

36°C Sunny

Search

16:46 07-09-2026